BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, SAHIWAL

INTERMEDIATE (12TH CLASS) EXAMINATION, 2022

CHEMISTRY

Past Paper 2022 • Paper Code: 8483

Retyped faithfully from the original board paper — same questions, no changes.

GROUP – I

Chemistry - Paper: II - Objective (ii)

Time Allowed: 20 Minutes | **Maximum Marks:** 17 | **Paper Code:** 8483

Note: Note: - You have four choices for each objective type question as A, B, C and D. The choice which you think is correct; fill that circle in front of that question number in your answer book. Use marker or pen to fill the circles. Cutting or filling up two or more circles will result no mark.

Note: SECTION-A

Q1. Which woody raw material is used for the manufacture of paper pulp?

(A) Cotton
(B) Bagasse
(C) Poplar
(D) Rice straw

Q2. Ecosystem is a smaller unit of:

(A) Lithosphere
(B) Hydrosphere
(C) Atmosphere
(D) Biosphere

Q3. A single chloride free radical can destroy how many Ozone molecules:

(A) 100
(B) 100000
(C) 10000
(D) 10

Q4. The correct statement is:

(A) Metallic character increases down the group
(B) Metallic character increases from left to right along a period
(C) Metallic character remains same from left to right along a period
(D) Metallic character remains same down the group

Q5. Chile saltpetre has the chemical formula:

(A) $NaNO_3$
(B) KNO_2
(C) $Na_2B_4O_7$
(D) $Na_2CO_3.H_2O$

Q6. The chief ore of Aluminum is:

(A) Na_3AlF_6
(B) $Al_2O_3.2H_2O$
(C) Al_2O_3
(D) $Al_2O_3.H_2O$

Q7. Ketones are prepared by the oxidation of:

(A) Primary alcohol
(B) Secondary alcohol
(C) Tertiary alcohol
(D) All of these

Q8. A carboxylic acid contains:

(A) A hydroxyl group
(B) A carboxyl group
(C) A hydroxyl and carboxyl group
(D) A phenolic group

Q9. Which of these polymers is synthetic polymer?

(A) Animal fat
(B) Starch
(C) Cellulose
(D) Polyester

Q10. Among the group VA elements, the most electronegative element is:

(A) Sb
(B) N
(C) P
(D) As

Q11. Chlorine heptaoxide (Cl_2O_7) reacts with water to form:

(A) Hypochlorous acid
(B) Chloric acid
(C) Perchloric acid
(D) Chlorine and Oxygen

Q12. Which of the given is a typical transition metal?

(A) Sc
(B) Y
(C) Ra
(D) Co

Q13. In, t-butyl alcohol, the tertiary carbon is bonded to:

(A) Two hydrogen atoms
(B) One hydrogen atom
(C) Three hydrogen atoms
(D) No hydrogen atom

Q14. Formula of Chloroform is:

(A) CH_3Cl
(B) CCl_4
(C) CH_2Cl_2
(D) $CHCl_3$

Q15. Aromatic hydrocarbons are the derivatives of:

(A) Normal series of paraffins
(B) Alkene
(C) Benzene
(D) Cyclohexane

Q16. When CO_2 is made to react with Ethyl Magnesium Iodide, followed by acid hydrolysis, the product formed is:

(A) Propane
(B) Propanoic acid
(C) Propanal
(D) Propanol

Q17. Which compound will have maximum repulsion with H_2O?

(A) C_6H_6
(B) CH_3-CH_2-OH
(C) CH_3-CH_2-CH_2-OH
(D) CH_3-O-CH_3

Chemistry - Paper: II - Subjective

Time Allowed: 2:40 Hours | **Maximum Marks:** 68

Note: Note: Section B is compulsory. Attempt any 3 questions from Section C.

Note: SECTION-B

Q2.. (Marks 16) Write short answers to any Eight parts. (8 x 2 = 16)

i. Why I.E of inert gases is maximum along the period?

ii. The hydration energies of different ions are in the given order $Al^{3+} > Mg^{2+} > Na^{+}$ Give reason.

iii. Why aq. solution of sodium carbonate is alkaline in nature?

iv. What is milk of Magnesia? Give its use as medicine.

v. SiO_2 exists in solid state while CO_2 in gaseous state. Give reason.

vi. Give four uses of Sodium Silicates.

vii. Give two reactions in which nitrous acid (HNO_2) behaves as oxidizing agent.

viii. What is Aqua Regia, how does it dissolve gold?

ix. What is Stadeler's process? Give its reaction.

x. What is Sacrificial corrosion?

xi. Define micronutrients. Give examples.

xii. What types of reactions taking place in first 24 hours during setting of cement?

Q3.. (Marks 16) Write short answers to any Eight parts. (8 x 2 = 16)

i. Describe the term "available chlorine" with reactions.

ii. How Bleaching powder is prepared by Hasenclever's Method?

iii. What are homocyclic and heterocyclic compounds. Give one example of each.

iv. Write down the structural formulas of (i) Naphthalene (ii) Anthracene.

v. How will you synthesize the following compounds starting from ethyne? (i) Oxalic acid (ii) Acrylonitrile.

vi. How does propyne react with the given reagents? (i) $AgNO_3/NH_4OH$ (ii) Cu_2Cl_2/NH_4OH

vii. Identify A and B. $CH_3CH_2CH_2OH$ with PCl_3 gives A; A with Alc. Ether gives B.

viii. Give IUPAC names of the given compounds: (a) CH_2Cl_2 (b) $BrCH_2\text{-}CH_2Br$

ix. How will you convert? $CH_3\text{-}CH_2\text{-}Br$ with Mg/Ether gives $CH_3COOC_2H_5$

x. What is meant by Rancidity of fats and oils?

xi. Demonstrate Hardening of Oils with example.

xii. Describe condensation polymerization with example.

Q4.. (Marks 12) Write short answers to any Six parts. (6 x 2 = 12)

i. Give names and possible isomeric structures of xylenes.

ii. Why benzene is less reactive than alkenes?

iii. Absolute alcohol cannot be prepared by fermentation process, Why?

iv. How would you convert phenol into cyclohexanol and picric acid?

v. Write down any four uses of formaldehyde.

vi. How carboxylic acids can be prepared by the hydrolysis of esters?

vii. Describe the synthesis of amino acid by Strecker method.

viii. Write down necessary conditions for the formation of smog.

ix. How detergents are threat to aquatic animal life?

> Note: SECTION-C (EACH QUESTION CARRIES EIGHT (8) MARKS)

Q5.. (Marks 4) (a) Define Ionization energy. How does it vary in groups and periods in the periodic table? (b) Define semi-conductor. Write two properties and two uses of semiconductor.

Q6.. (Marks 4) (a) Describe the commercial preparation of Sodium by Down's cell. (b) What is corrosion? Explain the electrochemical theory of corrosion.

Q7.. (Marks 4) (a) Describe any four features of organic compounds. 1x4 (b) Define alkyl halides, how are they prepared from alcohols. 1+3

Q8.. (Marks 4) (a) Define polymerisation. Explain the linear polymerisation of ethyne upto the formation of Neoprene. (b) How does acetaldehyde react with? (i) HCN (ii) $NaHSO_3$ (iii) NH_2OH (iv) H_2NNH_2

Q9.. (Marks 4) (a) Give the reaction of phenol with (i) HCHO (ii) Br_2 (iii) H_2/Ni 2+1+1 (b) Give the reaction mechanism of: (i) Nitration of benzene (ii) Sulphonation of benzene. 2+2

GROUP – II

Chemistry - Paper: II - Objective (ii)

Time Allowed: 20 Minutes | **Maximum Marks:** 17 | **Paper Code:** 8484

Note: Note: - You have four choices for each objective type question as A, B, C and D. The choice which you think is correct; fill that circle in front of that question number in your answer book. Use marker or pen to fill the circles. Cutting or filling up two or more circles will result no mark.

Note: SECTION-A

Q1. Phosphorus helps the growth of:

(A) Roots
(B) Leaves
(C) Stem
(D) Seed

Q2. The proportion of N_2 in atmosphere is:

(A) 21%
(B) 78%
(C) 0.9%
(D) 0.03%

Q3. The amount of Ozone in the atmosphere is expressed in unit:

(A) Debye
(B) Dobson
(C) a.m.u
(D) mg

Q4. Keeping in view the size of atom which order is correct one:

(A) Mg > Sr
(B) Ba > Mg
(C) Lu > Ce
(D) Cl > I

Q5. Which of the given sulphate is not soluble in water?

(A) Sodium Sulphate
(B) Potassium Sulphate
(C) Zinc Sulphate
(D) Barium Sulphate

Q6. Tincal is a mineral of:

(A) Al
(B) B
(C) Si
(D) C

Q7. The carbon atom of carbonyl group is:

(A) sp hybridized
(B) sp^2 hybridized
(C) sp^3 hybridized
(D) none of these

Q8. Which reagent is used to reduce a carboxylic group to an alcohol?

(A) H_2/Ni
(B) H_2/Pt
(C) $NaBH_4$
(D) $LiAlH_4$

Q9. Which of these polymers is a synthetic polymer?

(A) Animal fat
(B) Starch
(C) Cellulose
(D) Polyester

Q10. Oxidation of "NO" in air produces:

(A) N_2O
(B) N_2O_3
(C) N_2O_4
(D) N_2O_5

Q11. Which is the strongest acid?

(A) HClO
(B) $HClO_2$
(C) $HClO_3$
(D) $HClO_4$

Q12. Total number of transition metal (elements) is:

(A) 10
(B) 14
(C) 40
(D) 68

Q13. A double bond consists of:

(A) Two sigma bonds
(B) One sigma and one Pi bond
(C) One sigma and two Pi bonds
(D) Two Pi bonds

Q14. Vinyl acetylene combines with HCl to form

(A) Polyacetylene
(B) Benzene
(C) Chloroprene
(D) Divinyl acetylene

Q15. The benzene molecule contains:

(A) Three double bonds
(B) Two double bonds
(C) One double bond
(D) Delocalized π-electron charge

Q16. Which one of the given is not a nucleophile?

(A) H_2O (B) H_2S

(C) BF_3 (D) NH_3

Q17. Ethanol can be converted into ethanoic acid by:

(A) Hydrogenation (B) Hydration

(C) Oxidation (D) Fermentation

Chemistry - Paper: II - Subjective

Time Allowed: 2:40 Hours | **Maximum Marks:** 68

Note: Note: Section B is compulsory. Attempt any 3 questions from Section C.

Note: SECTION-B

Q2.. (Marks 16) Write short answers to any Eight parts. (8 x 2 = 16)

i. Justify by chemical reactions that ZnO is amphoteric in nature.

ii. Describe variation of melting point and boiling point in groups.

iii. Why CO_2 is gas but SiO_2 is solid?

iv. Why elements of group IA are also known as alkali metals?

v. Write chemical formula of Fluorite and Barite.

vi. Outline four principal uses of Borax.

vii. Nitrogen oxide is oxidizing agent, prove it by two reactions.

viii. How does dilute HNO_3 reacts with (i) Mn (ii) Mg

ix. Draw structure of CrO_4^{2-} and CrO_7^{2-} ions.

x. Define Chelates and give an example.

xi. Write names of four different steps involved in cement preparation.

xii. Define paper and fertilizer.

Q3.. (Marks 16) Write short answers to any Eight parts. (8 x 2 = 16)

i. Why HF is weaker acid than HCl?

ii. Write down uses of Argon.

iii. Define functional group isomerism. Give an example.

iv. Define Alicyclic compounds. Give two examples.

v. How Raney Nickel is prepared?

vi. Write down four uses of Ethene.

vii. Give the reactions of Ethyne with: (i) NH_3 (ii) HCN

viii. What is the excellent method for the preparation of Alkyl Iodide?

ix. Give the reaction of Grignard reagent with CO_2.

x. Write down four points of importance of Lipids.

xi. Define Iodine number and Acid number.

xii. How polyvinyl chloride (PVC) is prepared. Give its uses.

Q4.. (Marks 12) Write short answers to any Six parts. (6 x 2 = 12)

i. Give the reaction of side chain halogenation of benzene.

ii. Why nitration speeds up in the presence of H_2SO_4? Discuss.

iii. Give uses of Alcohol. (Any four)

iv. How does CH_3OH and C_2H_5OH react with I_2/NaOH? Give reaction.

v. Complete the reaction $CH_2 = CH_2 + O_2$ with $PdCl_2/CuCl_2/H_2O$

vi. Write any four uses of acetic acid.

vii. What are Essential and Non-essential amino acids? Discuss

viii. How Leather tanneries affect the quality of water? Discuss.

ix. Give the role of CFCs in ozone layer depletion? Give reaction.

> Note: SECTION-C (EACH QUESTION CARRIES EIGHT (8) MARKS)

Q5.. (Marks 4) (a) Define Oxides. Explain acidic and basic oxides. (b) Write note on applications of Aluminium (Any eight)

Q6.. (Marks 4) (a) Define Corrosion. Explain Electrochemical theory of Corrosion. (b) Discuss the commercial preparation of Caustic Soda by Diaphragm Cell. (Problems are not required).

Q7.. (Marks 4) (a) Define Cracking of Petroleum with any two types. (b) Explain reactivity of Alkyl halides on the basis of (i) Bond Energy. (ii) Bond Polarity.

Q8.. (Marks 4) (a) What is Raney Nickel? Give two examples in which it is used as catalyst. 1+3 (b) Give detail of Haloform reaction. Why it is called so? 3+1

Q9.. (Marks 4) (a) Explain nitration of benzene along with its mechanism. (b) How does ethyl alcohol react with: (i) Na (ii) PCl_5 (iii) NH_3 (iv) CH_3COOH.